

Exercise for Chapter 13

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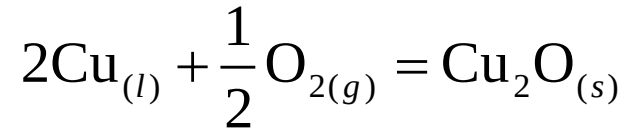
Chapter 13

13.1 Air at atmospheric pressure is blown over a Cu-rich copper–gold liquid solution at 1500 K. If only the copper is oxidized (to form pure solid Cu_2O), calculate the minimum activity of Cu which can be obtained in the solution.

Chapter 13

Solution to 13.1

For



$$\Delta G^\circ = -188,300 + 88.48T \text{ J}$$

$$\Delta G_{1500 \text{ K}}^\circ = -55,580 \text{ J} = RT \ln a_{\text{Cu}}^2 p_{\text{O}_2}^{1/2}$$

So

$$a_{\text{Cu}}^2 (0.21)^{1/2} = 0.0116$$

$$a_{\text{Cu}}^2 = 0.0253$$

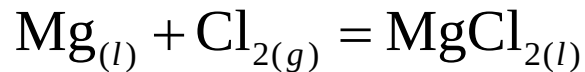
$$a_{\text{Cu}} = 0.159$$

Chapter 13

13.2 Magnesium can be removed from Mg–Al liquid solution by selectively forming the chloride MgCl_2 . Calculate the activity of Mg in the liquid Mg–Al system which can be achieved at 800 °C by reacting the solution with an H_2 –HCl gas mixture containing hydrogen at essentially 1 atm pressure and $p_{\text{HCl}} = 10^{-5}$ atm to form pure liquid MgCl_2 .

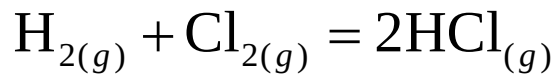
Chapter 13

Solution to 13.2



$$\Delta G^\circ = -603,200 + 121.42T \text{ J}$$

$$\Delta G_{1073 \text{ K}}^\circ = -472,916 \text{ J}$$



$$\Delta G^\circ = -188,200 - 12.80T \text{ J}$$

$$\Delta G_{1073 \text{ K}}^\circ = -201,933 \text{ J}$$

So

$$\text{Mg}_{(l)} + 2\text{HCl}_{(g)} = \text{MgCl}_{2(l)} + \text{H}_{2(g)} \quad \Delta G_{1073 \text{ K}}^\circ = -270,983 \text{ J} = RT \ln \frac{a_{\text{Mg}} \cdot p_{\text{HCl}}^2}{p_{\text{H}_2}}$$

$$\frac{a_{\text{Mg}} \cdot p_{\text{HCl}}^2}{p_{\text{H}_2}} = 6.43 \times 10^{-14} = \frac{a_{\text{Mg}} \cdot (10^{-5})^2}{1}$$

Which gives

$$a_{\text{Mg}} = 6.4 \times 10^{-4}$$

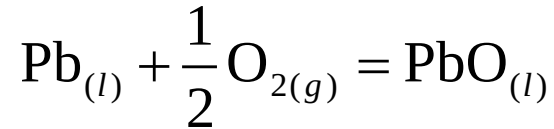
Chapter 13

13.3 The partial pressure of oxygen in equilibrium with pure liquid lead and pure liquid lead oxide at 1200 K is 2.16×10^{-9} atm. When SiO_2 is added to the liquid PbO to form a lead silicate melt, the oxygen pressure in equilibrium with pure liquid lead and the silicate melt is decreased to 5.41×10^{-10} atm. Calculate the activity of PbO in the lead silicate melt.

Chapter 13

Solution to 13.3

For



$$K_{1200 \text{ K}} = \frac{a_{\text{PbO}}}{a_{\text{Pb}} p_{\text{O}_2}^{1/2}} = \frac{1}{1 \times (2.16 \times 10^{-9})^{1/2}} = 21,517$$

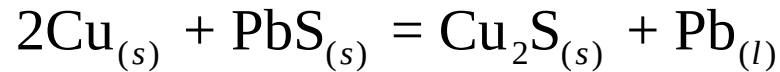
with

$$a_{\text{Pb}} = 1 \text{ atm, and } p_{\text{O}_2(\text{eq})} = 5.41 \times 10^{-10} \text{ atm}$$

$$a_{\text{PbO}} = 21,517 \times (5.41 \times 10^{-10})^{1/2} = 0.5$$

Chapter 13

13.4 Copper, present as an impurity in liquid Pb, can be removed by adding PbS to the Cu–Pb alloy and allowing the exchange reaction



to come to equilibrium.

The solid sulfides are mutually immiscible, Pb is insoluble in solid Cu, and the Cu liquidus, below 850 °C, can be represented by

$$\log X_{\text{Cu}} = -\frac{3500}{T} + 2.261$$

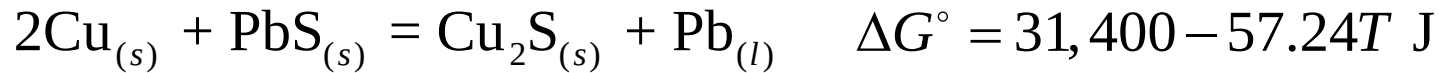
where X_{Cu} is the solubility of Cu in liquid Pb. If Cu obeys Henry's law in liquid Pb, calculate the extent to which Cu can be removed from liquid Pb by this process at 800 °C. Would the extent of purification of the lead be increased by increasing or by decreasing the temperature?

Chapter 13

Solution to 13.4



So for



$$\Delta G_{1073 \text{ K}}^\circ = -30,019 \text{ J} = -RT \ln K_{1073 \text{ K}}$$

Which gives

$$K_{1073 \text{ K}} = 28.93 = \frac{a_{\text{Pb}(l)}}{a_{\text{Cu}(s)}^2}$$

Chapter 13

Solution to 13.4

At 1073 K, $\log X_{\text{Cu}} = -\frac{3500}{T} + 2.261$ $X_{\text{Cu(liquidus)}} = 0.0998$

so $\gamma_{\text{Cu}} = \frac{1}{0.0998} = 10.02$

Thus

$$a_{\text{Cu}} = 10.02 X_{\text{Cu}} \quad \text{and} \quad a_{\text{Pb}} = 1 - X_{\text{Cu}}$$

$$K_{1073 \text{ K}} = 28.93 = \frac{1 - X_{\text{Cu}}}{100.4 X_{\text{Cu}}^2}$$

Which gives $X_{\text{Cu}} = 0.018$

At 850 °C, $X_{\text{Cu}} = 0.024$

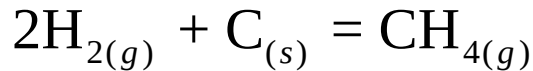
So extent of purification is increased by decreasing the temperature.

Chapter 13

13.5 A $\text{CH}_4\text{--H}_2$ gas mixture at 1 atm total pressure, in which $p_{\text{H}_2} = 0.957$ atm, is equilibrated with an Fe–C alloy at 1000 K. Calculate the activity of C with respect to graphite in the alloy. What would the value of p_{H_2} in the gas mixture (at $P_{\text{total}} = 1$ atm) have to be in order to saturate the Fe with graphite at 1000 K?

Chapter 13

Solution to 13.5



$$\Delta G^\circ = -91,040 + 110.7T \text{ J}$$

$$\Delta G_{1000 \text{ K}}^\circ = 19,660 \text{ J} = -RT \ln K_{1000 \text{ K}}$$

So

$$K_{1000 \text{ K}} = 9.40 \times 10^{-2} = \frac{p_{\text{CH}_4}}{a_{\text{C}} p_{\text{H}_2}^2} = \frac{0.043}{(0.957)^2 a_{\text{C}}}$$

Which gives $a_{\text{C}} = 0.5$

For $a_{\text{C}} = 1$ and $P_{\text{total}} = 1 \text{ atm}$

$$9.40 \times 10^{-2} = \frac{p_{\text{CH}_4}}{(1 - p_{\text{CH}_4})^2}$$

Which gives $p_{\text{H}_2} = 0.92 \text{ atm}$, $p_{\text{CH}_4} = 0.08 \text{ atm}$.

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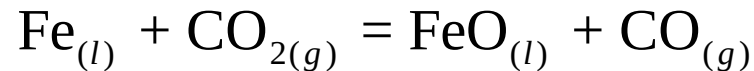
13.6 Calculate the activity of FeO in an FeO–Al₂O₃–SiO₂ melt below which the FeO cannot be reduced to pure liquid iron by a CO–CO₂ mixture of $p_{\text{CO}}/p_{\text{CO}_2} = 10^5$ at 1600 °C.

Chapter 13

Solution to 13.6

	$\Delta G_{1873\text{ K}}^{\circ}, \text{ J}$
$\text{FeO}(l)$	-155,557
$\text{CO}(g)$	-275,868
$\text{CO}_2(g)$	-395,673

So for



$$\Delta G_{1873\text{ K}}^{\circ} = -35,753 \text{ J} = -RT \ln K$$

Thus

$$K = 9.93 = \frac{a_{\text{FeO}} \cdot p_{\text{CO}}}{a_{\text{Fe}} \cdot p_{\text{CO}_2}} = a_{\text{FeO}} \times 10^5$$

Which gives $a_{\text{FeO}} = 9.9 \times 10^{-5}$

Chapter 13

- 13.7** A piece of iron is to be heat treated at 1000 K in a CO-CO₂-H₂O-H₂ gas mixture at 1 atm pressure. The gas mixture is produced by mixing CO₂ and H₂ and allowing the equilibrium $\text{CO}_2 + \text{H}_2 = \text{CO} + \text{H}_2\text{O}$ to establish. Calculate
- (a) The minimum H₂/CO₂ ratio in the inlet gas which can be admitted to the furnace without oxidizing the iron,
 - (b) the activity of carbon (with respect to graphite) in the equilibrated gas of this initial minimum H₂/CO₂ ratio,
 - (c) the total pressure to which the equilibrated gas would have to be raised to saturate the iron with graphite at 1000 K, and
 - (d) the effect, on the partial pressure of oxygen in the equilibrated gas, of this increase in total pressure.

Chapter 13

Solution to 13.7

(a)

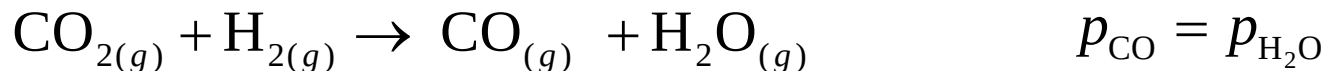
	$\Delta G_{1000\text{ K}}^{\circ}, \text{ J}$
CO	-199,350
CO ₂	-394,940
H ₂ O(g)	-191,650
FeO	-199,350

So

$$\text{CO}_{(g)} + \text{FeO}_{(s)} = \text{CO}_{2(g)} + \text{Fe}_{(s)} \quad \Delta G_{1000\text{ K}}^{\circ} = 3760 \text{ J} \quad K = \frac{p_{\text{CO}_2}}{p_{\text{CO}}} = 0.636$$

$$\text{H}_{2(g)} + \text{FeO}_{(s)} = \text{H}_2\text{O}_{(g)} + \text{Fe}_{(s)} \quad \Delta G_{1000\text{ K}}^{\circ} = 7700 \text{ J} \quad K = \frac{p_{\text{H}_2\text{O}}}{p_{\text{H}_2}} = 0.396$$

For



Chapter 13

Solution to 13.7

$$p_{\text{CO}} + p_{\text{CO}_2} + p_{\text{H}_2} + p_{\text{H}_2\text{O}} = 1 \text{ atm}$$

So

$$p_{\text{CO}} + 0.636 p_{\text{CO}} + \frac{p_{\text{H}_2\text{O}}}{0.396} + p_{\text{H}_2\text{O}} = 1 \text{ atm}$$

$$p_{\text{CO}} \left(1 + 0.636 + \frac{1}{0.396} + 1 \right) = 1 \text{ atm}$$

Which gives

$$p_{\text{CO}} = 0.1938 \text{ atm} \quad p_{\text{H}_2\text{O}} = 0.1938 \text{ atm}$$

$$p_{\text{CO}_2} = 0.1233 \text{ atm} \quad p_{\text{H}_2} = 0.4894 \text{ atm}$$

This gas is in equilibrium with Fe and FeO.

Chapter 13

Solution to 13.7

Mix CO_2 and H_2 in the ratio $\frac{\text{CO}_2}{\text{H}_2} = x$

	$\text{CO}_{2(g)}$	$+$	$\text{H}_{2(g)}$	$=$	$\text{CO}_{(g)}$	$+$	$\text{H}_2\text{O}_{(g)}$
before	x		1		0		0
after	$x-a$		$1-a$		a		a

Which gives

$$p_{\text{CO}} = \frac{a}{1+x} \quad p_{\text{H}_2\text{O}} = \frac{a}{1+x} \quad p_{\text{CO}_2} = \frac{x-a}{1+x} \quad p_{\text{H}_2} = \frac{1-a}{1+x}$$

For equilibrium with Fe and FeO:

$$\frac{p_{\text{CO}_2}}{p_{\text{CO}}} = 0.636 = \frac{x-a}{a} \quad \frac{p_{\text{H}_2\text{O}}}{p_{\text{H}_2}} = 0.396 = \frac{a}{1-a}$$

Which gives $a = 0.2837$ $x = 0.4641 = \frac{\text{CO}_2}{\text{H}_2} \Big|_{\text{initial}}$

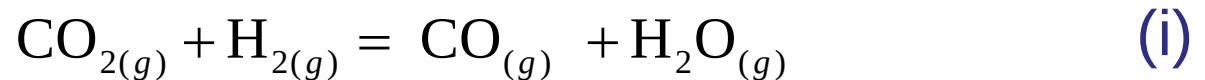
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Solution to 13.7

For $\text{Fe}_{(s)} + \frac{1}{2} \text{O}_{2(g)} = \text{FeO}_{(s)} \quad \Delta G_{1000 \text{ K}}^{\circ} = -199,350 \text{ J} = \frac{1}{2} RT \ln p_{\text{O}_2}$

So $p_{\text{O}_2(\text{Fe/FeO})} = 1.494 \times 10^{-21} \text{ atm}$

Choose a larger value of x, say $x = 1$, then



before $\quad 1 \quad 1 \quad 0 \quad 0$

after $\quad 1-a \quad 1-a \quad a \quad a$

So $p_{\text{CO}} = \frac{a}{2} = p_{\text{H}_2\text{O}}; \quad p_{\text{CO}_2} = \frac{1-a}{2} = p_{\text{H}_2}$

$$\Delta G_{(i), 1000 \text{ K}}^{\circ} = 3040 \text{ J} = -RT \ln \frac{p_{\text{CO}} \cdot p_{\text{H}_2\text{O}}}{p_{\text{CO}_2} \cdot p_{\text{H}_2}}$$

Thus $0.6226 = \frac{a^2}{(1-a)^2} \quad \text{which gives } a = 0.441$

Chapter 13

Solution to 13.7

So $p_{\text{CO}} = 0.2205 \text{ atm}$ $p_{\text{CO}_2} = 0.2795 \text{ atm}$ $\frac{p_{\text{CO}_2}}{p_{\text{CO}}} = 1.268$



$$K_{1000 \text{ K}} = 1.64 \times 10^{-10} = \frac{p_{\text{CO}_2}}{p_{\text{CO}} \cdot p_{\text{O}_2}^{1/2}} = \frac{1.268}{p_{\text{O}_2}^{1/2}}$$

Which gives $p_{\text{O}_2} = 5.93 \times 10^{-21} \text{ atm}$, which is greater than $p_{\text{O}_2(\text{eq, Fe/FeO})}$ at 1000 K.

So 0.4641 is the maximum allowable ratio of $\left. \frac{\text{CO}_2}{\text{H}_2} \right|_{\text{initial}}$

and 2.155 is the minimum allowable ratio of $\left. \frac{\text{H}_2}{\text{CO}_2} \right|_{\text{initial}}$

Chapter 13

Solution to 13.7

(b) $\text{C}_{(s)} + \text{CO}_{2(g)} = 2\text{CO}_{(g)} \quad \Delta G_{1000\text{ K}}^{\circ} = -3760 \text{ J} = -RT \ln \frac{p_{\text{CO}}^2}{p_{\text{CO}_2} \cdot a_{\text{C}}}$

So $\frac{p_{\text{CO}}^2}{p_{\text{CO}_2} \cdot a_{\text{C}}} = 1.572 = \frac{0.1938^2}{0.1233 \cdot a_{\text{C}}}$, which gives $a_{\text{C}} = 0.194$

(c) For $a_{\text{C}} = 1$ and pressure P ,

$$1.572 = \frac{0.1938^2 P}{0.1233}$$

Which gives $P = 5.16 \text{ atm}$.

(d) The values of the ratios $\frac{p_{\text{CO}_2}}{p_{\text{CO}}}$ and $\frac{p_{\text{H}_2\text{O}}}{p_{\text{H}_2}}$ are not functions of the total pressure P .

Chapter 13

13.8 An Fe–Mn solid solution containing $X_{\text{Mn}} = 0.001$ is in equilibrium with an FeO–MnO solid solution and an oxygen-containing gaseous atmosphere at 1000 K. How many degrees of freedom does the equilibrium have? What is the composition of the equilibrium oxide solution, and what is the partial pressure of oxygen in the gas phase? Assume that both solid solutions are Raoultian in their behavior.

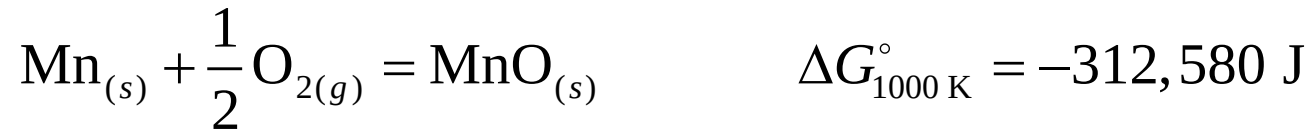
Chapter 13

Solution to 13.8

$C = 3$ (Fe, Mn, O), $P = 3$ (metal, oxide, gas)

So
$$F = C + 2 - P = 3 + 2 - 3 = 2$$

Fixing the temperature at 1000 K leaves 1 degree of freedom



So
$$K_{\text{Mn}} = 2.124 \times 10^{16} = \frac{X_{\text{MnO}}}{X_{\text{Mn}} \cdot p_{\text{O}_2}^{1/2}}$$



So
$$K_{\text{Fe}} = 2.587 \times 10^{10} = \frac{X_{\text{FeO}}}{X_{\text{Fe}} \cdot p_{\text{O}_2}^{1/2}}$$

Chapter 13

Solution to 13.8

$$\frac{K_{\text{Mn}}}{K_{\text{Fe}}} = 821,215 = \frac{X_{\text{MnO}} X_{\text{Fe}}}{X_{\text{Mn}} \cdot X_{\text{FeO}}} = \frac{(1 - X_{\text{FeO}}) \cdot 0.999}{0.001 \cdot X_{\text{FeO}}}$$

Which gives

$$X_{\text{FeO}} = 1.22 \times 10^{-3}$$

Then

$$K_{\text{Fe}} = 2.587 \times 10^{10} = \frac{1.22 \times 10^{-3}}{0.999 \cdot p_{\text{O}_2}^{1/2}}$$

gives

$$p_{\text{O}_2} = 2.23 \times 10^{-27} \text{ atm}$$

Chapter 13

13.9 The elements A and B, which are both solid at 1000 °C, form two stoichiometric compounds A_2B and AB_2 , which are also both solid at 1000 °C. The system A–B does not contain any solid solutions. A has an immeasurably small vapor pressure at 1000 °C, and, for the change of state $B_{(s)} = B_{(v)}$,

$$\Delta G^\circ = 187,200 - 108.80T \text{ J}$$

The vapor pressure exerted by an equilibrated AB_2 – A_2B mixture is given by

$$\log p(\text{atm}) = -\frac{11,242}{T} + 6.53$$

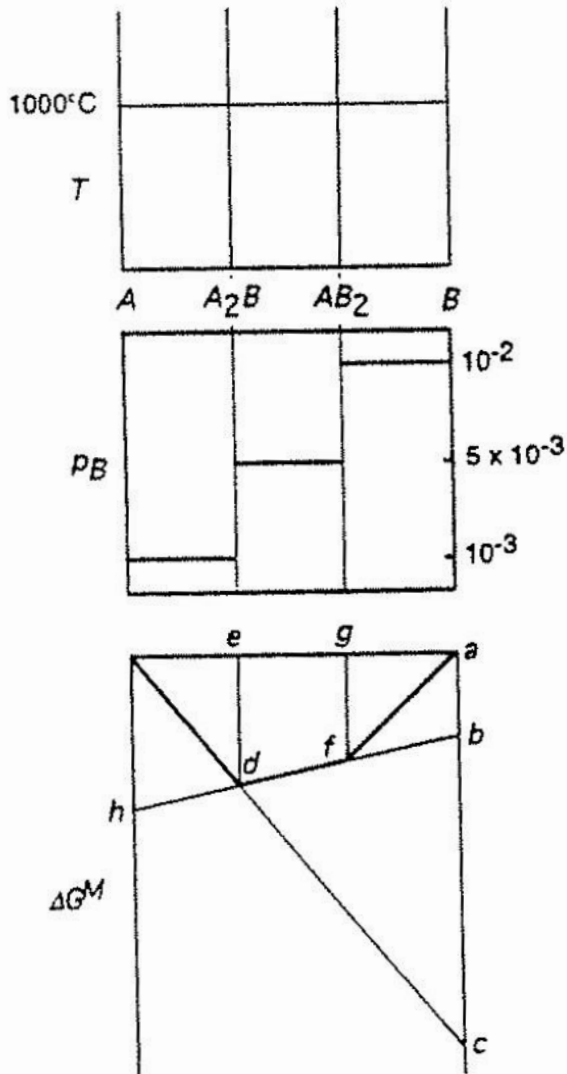
and the vapor pressure exerted by an equilibrated A– A_2B mixture is given by

$$\log p(\text{atm}) = -\frac{12,603}{T} + 6.9$$

From these data, calculate the standard Gibbs free energies of formation of A_2B and AB_2 .

Chapter 13

Solution to 13.9



$$a_B \text{ in } A_2B-AB_2 \text{ mixture} = \frac{5 \times 10^{-3}}{10^{-2}} = 0.5$$

$$\text{So } ab = RT \ln a_B = -7336 \text{ J}$$

$$a_B \text{ in } A-A_2B \text{ mixture} = \frac{10^{-3}}{10^{-2}} = 0.1$$

$$\text{So } ac = RT \ln a_B = -24,371 \text{ J}$$

$$\text{Thus } ed = \frac{-24,371}{3} = -8123 \text{ J}$$

$$\text{For } 2A + B = A_2B \quad \Delta G^\circ = 3 \times ed = -24,370 \text{ J}$$

$$gf = \frac{ed + ab}{2} = -7730 \text{ J}$$

$$\text{For } A + 2B = AB_2 \quad \Delta G^\circ = 3 \times gf = -23,190 \text{ J}$$

Chapter 13

13.10 For the change of standard state $V_{(s)} = V_{(1 \text{ wt\% in Fe})}$

$$\Delta G^\circ = -15,480 - 45.61T \text{ J}$$

Calculate the value of γ_V° at 1600 °C. If a liquid Fe–V solution is equilibrated with pure solid VO and a gas containing $p_{O_2} = 4.72 \times 10^{-10}$ atm, calculate the activity of V in the liquid solution

- (a) with respect to solid V as the standard state,
- (b) with respect to liquid V as the standard state,
- (c) with respect to the Henrian standard state, and
- (d) with respect to the 1 wt% in iron standard state.

Chapter 13

Solution to 13.10

$$V_{(s)} \rightarrow V_{(1 \text{ wt}\%)} \quad \Delta G_{1873 \text{ K}}^{\circ} = RT \ln \frac{\gamma_v^{\circ} 55.85}{100 \times 50.94}$$

So $\gamma_v^{\circ} = 0.14$

$$V_{(s)} + \frac{1}{2} O_2 = VO_{(s)} \quad \Delta G_{1873 \text{ K}}^{\circ} = -274,785 = -RT \ln \frac{1}{a_{V(s)} [4.72 \times 10^{-10}]^{1/2}}$$

So $a_{V(s)} = 10^{-3}$

$$V_{(s)} \rightarrow V_{(l)} \quad \Delta G^{\circ} = 22,840 - \frac{22,840}{2193} 1873 = 3333 \text{ J}$$

So $V_{(l)} + \frac{1}{2} O_2 = VO_{(s)}$

$$\Delta G_{1873 \text{ K}}^{\circ} = -274,785 - 3333 = -278,118 \text{ J} = -RT \ln \frac{1}{a_{V(l)} [4.72 \times 10^{-10}]^{1/2}}$$

Thus $a_{V(l)} = 8.07 \times 10^{-4}$

Chapter 13

Solution to 13.10

$$a_{V(s)} = h_V \cdot \gamma_v^\circ$$

So
$$h_V = \frac{10^{-3}}{0.14} = 7.13 \times 10^{-3}$$

$$\frac{h_{V(1\%)}}{h_V} = \frac{50.95 \times 100}{55.85} = 91.24$$

So

$$h_{V(1\%)} = 91.24 \times 7.13 \times 10^{-3} = 0.65$$

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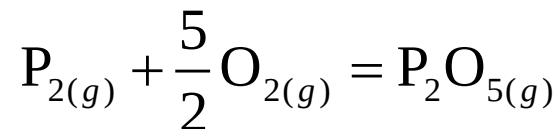
13.11 When an Fe–P liquid solution is equilibrated at 1900 K with solid CaO, solid $3\text{CaO} \cdot \text{P}_2\text{O}_5$, and a gas phase containing $p_{\text{O}_2} = 10^{-10}$ atm, the activity of P in the iron, with respect to the 1 wt% in Fe standard state, is 20. Given that $\Delta G_{1900\text{ K}}^\circ = -564,600$ J for

$$3\text{CaO}_{(s)} + \text{P}_2\text{O}_{5(g)} = 3\text{CaO} \cdot \text{P}_2\text{O}_{5(s)}$$

and $\Delta G^\circ = -122,200 - 19.22T$ J for

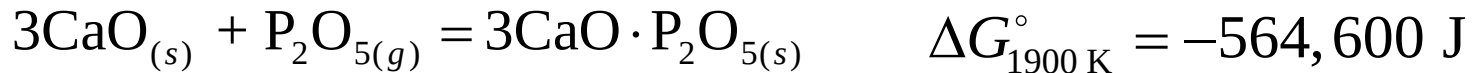
$$\frac{1}{2}\text{P}_{2(g)} = \text{P}_{(1\text{ wt\% in Fe})}$$

calculate $\Delta G_{1900\text{ K}}^\circ$ for the reaction



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Solution to 13.11



So $K = 3.325 \times 10^{15} = \frac{1}{p_{\text{P}_2\text{O}_5}} \quad p_{\text{P}_2\text{O}_5} = 3 \times 10^{-16} \text{ atm}$

$$\frac{1}{2} \text{P}_{2(g)} = \text{P}_{(1 \text{ wt\% in Fe})} \quad \Delta G^{\circ} = -122,200 - 19.22T \text{ J} \quad \Delta G_{1900\text{ K}}^{\circ} = -158,680 \text{ J}$$

So $K = 2.303 \times 10^4 = \frac{h_{\text{P}(1 \text{ wt\%})}}{p_{\text{P}_2}^{1/2}} = \frac{20}{p_{\text{P}_2}^{1/2}} \quad p_{\text{P}_2} = 7.539 \times 10^{-7} \text{ atm}$

Thus for

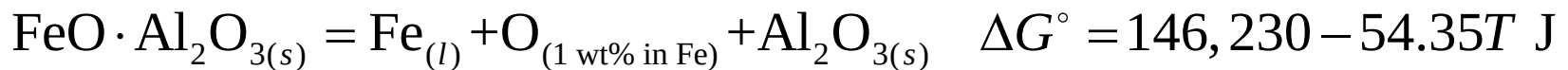
$$\text{P}_{2(g)} + \frac{5}{2} \text{O}_{2(g)} = \text{P}_2\text{O}_{5(g)}$$
$$K = \frac{3 \times 10^{-16}}{7.539 \times 10^{-7} \times (10^{-10})^{5/2}} = 3.99 \times 10^{15}$$

$$\Delta G_{1900\text{ K}}^{\circ} = -RT \ln K = -567,500 \text{ J}$$

Chapter 13

13.12 Liquid iron, contained in an Al_2O_3 crucible under a gaseous atmosphere of $p_{\text{O}_2} = 3 \times 10^{-12}$ atm at 1600 °C, contains its equilibrium contents of dissolved oxygen and aluminum. To what value must p_{O_2} be raised in order that solid hercynite ($\text{FeO} \cdot \text{Al}_2\text{O}_3$) appears in equilibrium with the melt and with solid Al_2O_3 ? What is the activity of Al (with respect to the 1 wt% in Fe standard state) in this state? How many degrees of freedom does this equilibrium have at 1600 °C?

Given



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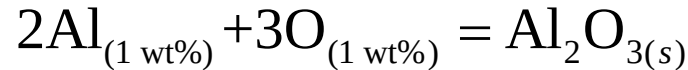
Solution to 13.12

$$\text{Al}_{(l)} = \text{Al}_{(1 \text{ wt\% in Fe})} \quad \Delta G^\circ = -43,100 - 32.26T \text{ J} \quad \Delta G_{1873 \text{ K}}^\circ = 103,523 \text{ J}$$

$$\text{O}_{(1 \text{ wt\% in Fe})} = \frac{1}{2} \text{O}_{2(g)} \quad \Delta G^\circ = 111,070 + 5.87T \text{ J} \quad \Delta G_{1873 \text{ K}}^\circ = 122,064 \text{ J} \quad (\text{i})$$

$$2\text{Al}_{(l)} + \frac{3}{2} \text{O}_{2(g)} = \text{Al}_2\text{O}_{3(s)} \quad \Delta G_{1873 \text{ K}}^\circ = -1,075,103 \text{ J}$$

So



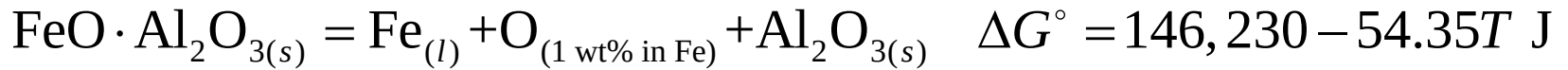
$$\Delta G_{1873 \text{ K}}^\circ = -501,865 \text{ J} = 8.3144 \times 1873 \times 2.303 \log [\text{O}]^3 [\text{Al}]^2$$

For saturation with Al_2O_3

$$2 \log [\text{Al}] + 3 \log [\text{O}] = -14 \quad (\text{ii})$$

Chapter 13

Solution to 13.12



$$\Delta G_{1873 \text{ K}}^\circ = 44,432 \text{ J} = -RT \ln [\text{O}]$$

$$[\text{O}] = 5.77 \times 10^{-2} \text{ wt\%}$$

From (i)

$$122,064 = -RT \ln \frac{p_{\text{O}_2}^{1/2}}{[\text{O}]}$$

So $\frac{p_{\text{O}_2}^{1/2}}{[\text{O}]} = 3.944 \times 10^{-4} \quad p_{\text{O}_2} = 5.17 \times 10^{-10} \text{ atm}$

And from (ii)

$$[\text{Al}] = 7.215 \times 10^{-6} \text{ wt\%}$$

$$C = 3 (\text{Al, Fe, O}); \quad P = 4 (\text{metal, Al}_2\text{O}_3, \text{FeO} \cdot \text{Al}_2\text{O}_3, \text{gas})$$

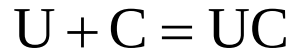
So $F = C + 2 - P = 1$, which is specified by selecting $T = 1600^\circ\text{C}$.

Chapter 13

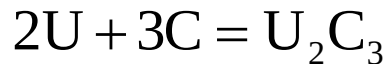
13.13 UC_2 can be equilibrated with UC and C at high temperature and can be equilibrated with U_2C_3 and C at lower temperatures. Calculate the maximum and minimum temperatures at which UC_2 can exist.

Chapter 13

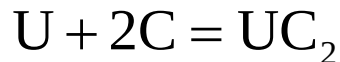
Solution to 13.13



$$\Delta G^\circ = -102,900 + 5.02T \text{ J}$$



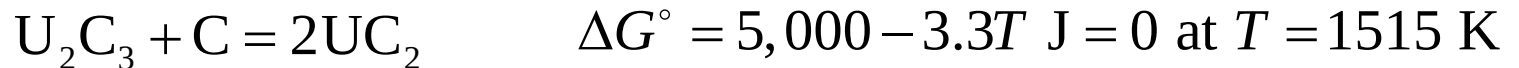
$$\Delta G^\circ = -236,800 + 25.1T \text{ J}$$



$$\Delta G^\circ = -115,900 + 10.9T \text{ J}$$

So $\text{UC} + \text{C} = \text{UC}_2$ $\Delta G^\circ = -13,000 + 5.88T \text{ J} = 0$ at $T = 2211 \text{ K}$

At $T > 2211 \text{ K}$, UC_2 metastable w.r.t $\text{UC} + \text{C}$



Thus at $T < 1515 \text{ K}$, UC_2 metastable w.r.t $\text{U}_2\text{C}_3 + \text{C}$

So $T_{\text{max}} = 2211 \text{ K}$

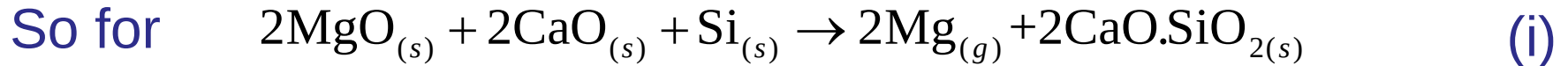
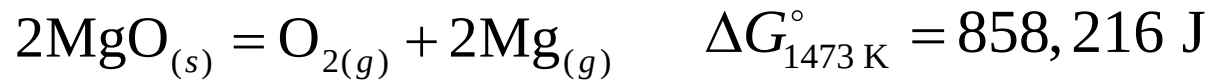
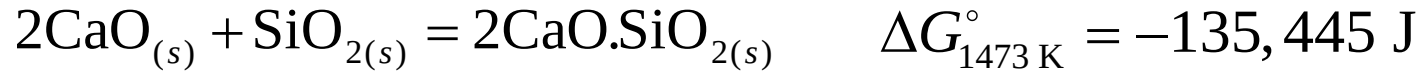
$T_{\text{min}} = 1515 \text{ K}$

Chapter 13

13.14 In the Pigeon process for the production of magnesium, dolomite ($\text{CaO} \cdot \text{MgO}$) is reduced by silicon to form magnesium vapor and $2\text{CaO} \cdot \text{SiO}_2$. Calculate the equilibrium pressure of magnesium vapor produced by this reaction at 1200°C . The Gibbs free energy of formation of dolomite from CaO and MgO is small enough that it can be ignored.

Chapter 13

Solution to 13.14



$$\Delta G_{(i),1473\text{ K}}^{\circ} = 71,973\text{ J} = -RT \ln p_{\text{Mg}}^2 = -8.3144 \times 1473 \ln p_{\text{Mg}}^2$$

Which gives

$$p_{\text{Mg}} = 0.053\text{ atm}$$

Chapter 13

13.15 What is the minimum value that the activity of MgO can have in $\text{MgO} \cdot \text{Al}_2\text{O}_3$ at 1000 °C?

Chapter 13

Solution to 13.15

For

$$\text{MgO}_{(s)} + \text{Al}_2\text{O}_{3(s)} = \text{MgO} \cdot \text{Al}_2\text{O}_{3(s)}$$
$$\Delta G_{1273\text{ K}}^{\circ} = -38,220\text{ J} = -RT \ln \frac{a_{\text{MgO} \cdot \text{Al}_2\text{O}_3}}{a_{\text{MgO}} a_{\text{Al}_2\text{O}_3}}$$

The minimum activity of MgO in $\text{MgO} \cdot \text{Al}_2\text{O}_3$ occur when it is saturated with Al_2O_3 at $a_{\text{Al}_2\text{O}_3} = 1$.

So with $a_{\text{MgO} \cdot \text{Al}_2\text{O}_3} = a_{\text{Al}_2\text{O}_3} = 1$

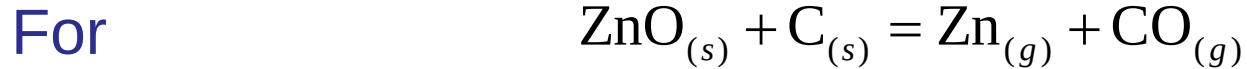
$$a_{\text{MgO}} = 0.027$$

Chapter 13

13.16 A mixture of ZnO and graphite is placed in an evacuated vessel and heated to 1200 K. Calculate the partial pressures of Zn, CO, and CO₂ that are developed.

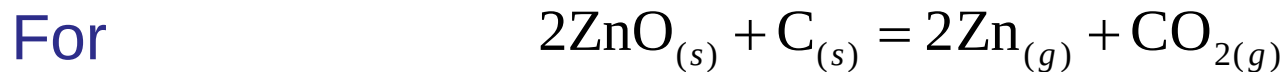
Chapter 13

Solution to 13.16



$$\Delta G_{1200\text{ K}}^{\circ} = 5720\text{ J} = -RT \ln p_{\text{Zn}} \cdot p_{\text{CO}}$$

So
$$p_{\text{Zn}} \cdot p_{\text{CO}} = 0.564$$



$$\Delta G_{1200\text{ K}}^{\circ} = 50,092\text{ J} = -RT \ln p_{\text{Zn}}^2 \cdot p_{\text{CO}_2}$$

So
$$p_{\text{Zn}}^2 \cdot p_{\text{CO}_2} = 6.60 \times 10^{-3}$$

Chapter 13

Solution to 13.16

In the gas phase

$$\frac{n_{\text{Zn}}}{n_{\text{O}}} = 1 = \frac{p_{\text{Zn}}}{p_{\text{CO}} + 2p_{\text{CO}_2}}$$

So

$$p_{\text{Zn}} = p_{\text{CO}} + 2p_{\text{CO}_2}$$

Thus

$$p_{\text{CO}} (p_{\text{CO}} + 2p_{\text{CO}_2}) = 0.564$$

$$p_{\text{CO}_2} (p_{\text{CO}} + 2p_{\text{CO}_2})^2 = 6.60 \times 10^{-3}$$

Which gives

$$p_{\text{CO}} = 0.740 \text{ atm,}$$

$$p_{\text{CO}_2} = 0.0114 \text{ atm,}$$

$$p_{\text{Zn}} = 0.763 \text{ atm,}$$

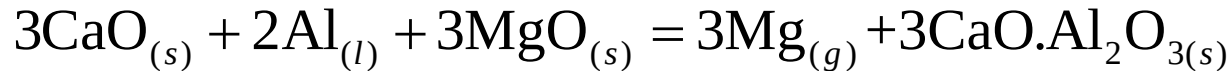
Chapter 13

13.17 An assemblage of solid CaO , MgO , $3\text{CaO} \cdot \text{Al}_2\text{O}_3$, and liquid Al exerts an equilibrium vapor pressure of Mg of 0.035 atm at 1300 K. Write the equation for the appropriate reaction equilibrium. Calculate the standard Gibbs free energy of formation of $3\text{CaO} \cdot \text{Al}_2\text{O}_3$ from CaO and Al_2O_3 and the activity of Al_2O_3 in CaO -saturated $3\text{CaO} \cdot \text{Al}_2\text{O}_3$ at 1300 K.

Chapter 13

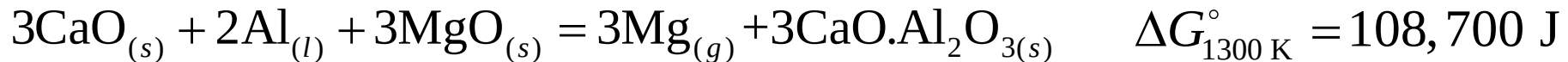
Solution to 13.17

The equilibrium is



for which $K = p_{\text{Mg}}^3 = 0.035^3 = 4.288 \times 10^{-5}$

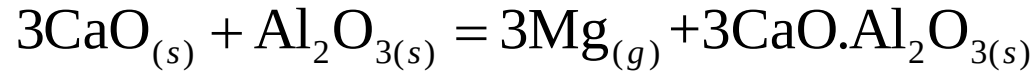
So $\Delta G_{1300\text{ K}}^\circ = -8.3144 \times 1300 \times \ln(4.288 \times 10^{-5}) \text{ J} = 108,700 \text{ J}$



Chapter 13

Solution to 13.17

Thus for



$$\Delta G_{1300\text{ K}}^{\circ} = -22,140\text{ J}$$

with $a_{\text{CaO}.\text{Al}_2\text{O}_3} = a_{\text{CaO}} = 1$

$$-22,140 = -RT \ln \frac{1}{a_{\text{Al}_2\text{O}_3}}$$

Which gives

$$a_{\text{Al}_2\text{O}_3} = 0.129$$

Chapter 13

13.18 An iron–carbon melt containing 0.5 wt% C is prepared in an alumina crucible under an atmosphere of $p_{\text{CO}} = 1$ atm at 1600 °C. Calculate the equilibrium concentrations of O and Al in the melt

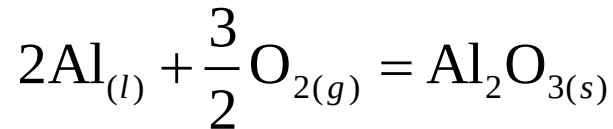
- (a) ignoring all solute–solute interactions and
- (b) considering the solute–solute interactions.

The interaction coefficients are listed in Table 13.1.

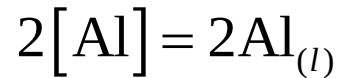
Chapter 13

Solution to 13.18

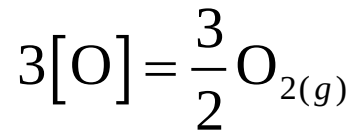
(a)



$$\Delta G_{1873\text{ K}}^{\circ} = -1,075,103\text{ J}$$



$$\Delta G_{1873\text{ K}}^{\circ} = 267,045\text{ J}$$



$$\Delta G_{1873\text{ K}}^{\circ} = 366,193\text{ J}$$

So for

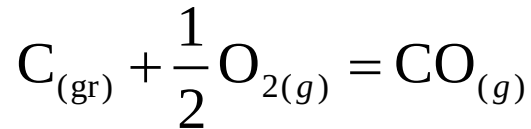
$$2[\text{Al}] + 3[\text{O}] = \text{Al}_2\text{O}_{3(s)} \quad \Delta G_{1873\text{ K}}^{\circ} = -501,864\text{ J} = RT \ln [\%\text{Al}]^2 [\%\text{O}]^3$$

Which gives

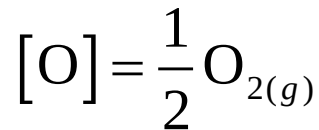
$$2\log [\%\text{Al}] + 3\log [\%\text{O}] = -13.996 \quad (\text{i})$$

Chapter 13

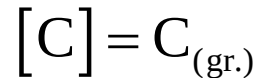
Solution to 13.18



$$\Delta G_{1873 \text{ K}}^{\circ} = -275,868 \text{ J}$$



$$\Delta G_{1873 \text{ K}}^{\circ} = 122,064 \text{ J}$$



$$\Delta G_{1873 \text{ K}}^{\circ} = 56,552 \text{ J}$$

So for

$$[\text{C}] + [\text{O}] = \text{CO}_{(\text{g})} \quad \Delta G_{1873 \text{ K}}^{\circ} = -97,250 \text{ J} = -RT \ln \frac{p_{\text{CO}}}{[\% \text{C}][\% \text{O}]}$$

with $p_{\text{CO}} = 1 \text{ atm}$,

$$\log [\% \text{C}] + \log [\% \text{O}] = -2.716 \quad (\text{ii})$$

$$\text{So } \log [\% \text{O}] = -2.716 - \log 0.5 = -2.414 \quad \text{and} \quad [\% \text{O}] = 0.00385$$

$$\log [\% \text{Al}] = \frac{-13.996 - 3(-2.414)}{2} = -3.377 \quad \text{and} \quad [\% \text{Al}] = 0.00042$$

Chapter 13

Solution to 13.18

(b)

Eq.(i) becomes $2\log[h_{\text{Al}}] + 3\log[h_{\text{O}}] = -13.996$

Eq.(ii) becomes $\log[h_{\text{C}}] + \log[h_{\text{O}}] = -2.716$

$$\log[f_{\text{C}}] = e_{\text{C}}^{\text{C}}[\text{C}] + e_{\text{C}}^{\text{O}}[\text{O}] + e_{\text{C}}^{\text{Al}}[\text{Al}]$$

$$\log[f_{\text{O}}] = e_{\text{O}}^{\text{O}}[\text{O}] + e_{\text{O}}^{\text{C}}[\text{C}] + e_{\text{O}}^{\text{Al}}[\text{Al}]$$

$$\log[f_{\text{Al}}] = e_{\text{Al}}^{\text{Al}}[\text{Al}] + e_{\text{Al}}^{\text{C}}[\text{C}] + e_{\text{Al}}^{\text{O}}[\text{O}]$$

where $e_{\text{C}}^{\text{C}} = 22 \times 10^{-2}$, $e_{\text{O}}^{\text{O}} = -20 \times 10^{-2}$, $e_{\text{Al}}^{\text{Al}} = 4.8 \times 10^{-2}$

$$e_{\text{C}}^{\text{O}} = -9.7 \times 10^{-2}, \quad e_{\text{O}}^{\text{C}} = -13 \times 10^{-2}, \quad e_{\text{Al}}^{\text{C}} = 11 \times 10^{-2}$$

$$e_{\text{C}}^{\text{Al}} = 4.8 \times 10^{-2}, \quad e_{\text{O}}^{\text{Al}} = -94 \times 10^{-2}, \quad e_{\text{Al}}^{\text{O}} = -160 \times 10^{-2}$$

Chapter 13

Solution to 13.18

So

$$2\{\log f_{\text{Al}} + \log[\text{Al}]\} + 3\{\log f_{\text{O}} + \log[\text{O}]\} = -13.996$$

$$\begin{aligned} \text{or } 2\{e_{\text{Al}}^{\text{Al}}[\text{Al}] + e_{\text{Al}}^{\text{C}}[\text{C}] + e_{\text{Al}}^{\text{O}}[\text{O}] + \log[\text{Al}]\} \\ + 3\{e_{\text{O}}^{\text{O}}[\text{O}] + e_{\text{O}}^{\text{C}}[\text{C}] + e_{\text{O}}^{\text{Al}}[\text{Al}] + \log[\text{O}]\} = -13.996 \end{aligned}$$

$$\log f_{\text{C}} + \log[\text{C}] + \log f_{\text{O}} + \log[\text{O}] = -2.716$$

$$\begin{aligned} \text{or } e_{\text{C}}^{\text{C}}[\text{C}] + e_{\text{C}}^{\text{O}}[\text{O}] + e_{\text{C}}^{\text{Al}}[\text{Al}] + \log[\text{C}] \\ + e_{\text{O}}^{\text{O}}[\text{O}] + e_{\text{O}}^{\text{C}}[\text{C}] + e_{\text{O}}^{\text{Al}}[\text{Al}] + \log[\text{O}] = -2.716 \end{aligned}$$

Chapter 13

Solution to 13.18

Thus

$$2\{4.8 \times 10^{-2} [Al] + 11 \times 10^{-2} [C] - 160 \times 10^{-2} [O] + \log [Al]\} \\ + 3\{-20 \times 10^{-2} [O] - 13 \times 10^{-2} [C] - 94 \times 10^{-2} [Al] + \log [O]\} = -13.996$$

and

$$22 \times 10^{-2} [C] - 9.7 \times 10^{-2} [O] + 4.8 \times 10^{-2} [Al] + \log [C] \\ - 20 \times 10^{-2} [O] - 13 \times 10^{-2} [C] - 94 \times 10^{-2} [Al] + \log [O] = -2.716$$

with $[C] = 0.5$

$$\%Al = 0.000541$$

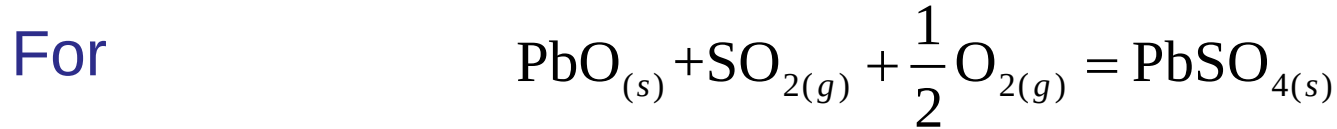
$$\%O = 0.00351$$

Chapter 13

13.19 It is required that PbO be eliminated from an ore containing PbO, PbS, and PbSO₄ by converting it to PbS or PbSO₄ by reaction with an SO₂–O₂ gas. Although the pressure of O₂ in the gas can vary within wide limits, the partial pressure of SO₂ may not be higher than 0.5 atm. Calculate the maximum temperature at which it can be guaranteed that the PbO phase will be eliminated.

Chapter 13

Solution to 13.19

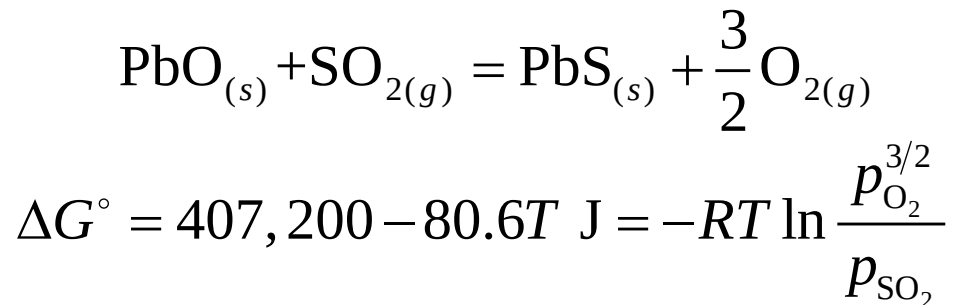


$$\Delta G^\circ = -401,200 + 261.5T \text{ J} = RT \ln p_{\text{SO}_2} \cdot p_{\text{O}_2}^{1/2}$$

So

$$\frac{-48253}{T} + 31.45 = \ln p_{\text{SO}_2} + \frac{1}{2} \ln p_{\text{O}_2} \quad (\text{i})$$

For



So

$$\frac{-48975}{T} + 9.69 = \frac{3}{2} \ln p_{\text{O}_2} - \ln p_{\text{SO}_2} \quad (\text{ii})$$

Chapter 13

Solution to 13.19

With $p_{\text{SO}_2} = 0.5 \text{ atm}$

(i) becomes

$$\ln p_{\text{O}_2} = \frac{-96506}{T} + 63.59 \quad (\text{iii})$$

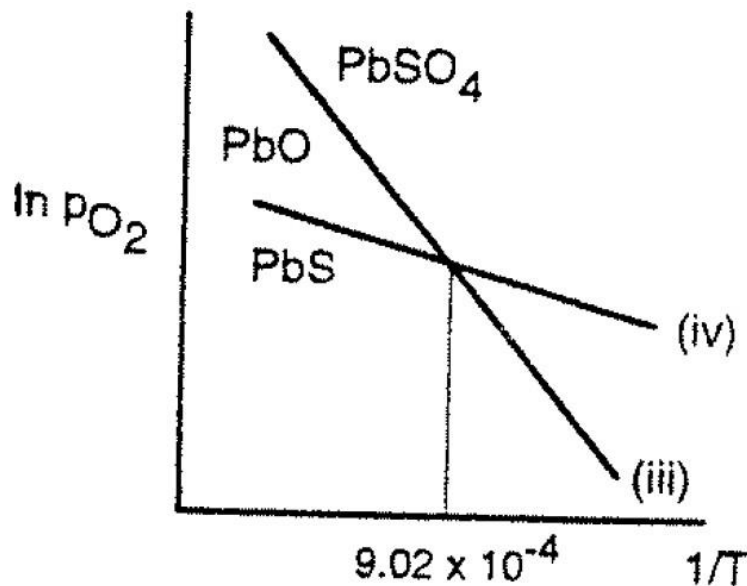
and (ii) becomes

$$\ln p_{\text{O}_2} = \frac{-32650}{T} + 5.998 \quad (\text{iv})$$

(iii) and (iv) intersect at

$$\frac{1}{T} = 9.02 \times 10^{-4}, \quad \ln p_{\text{O}_2} = -23.46$$

So PbO is not stable at $T > 1108 \text{ K}$, which is thus the maximum temperature at which it is eliminated.



Chapter 13

13.20 Cementite, Fe_3C , is metastable with respect to carbon-saturated α -iron and graphite at 950 K and 1 atm pressure. Given that the molar volumes of α -Fe, graphite, and Fe_3C at 950 K are, respectively, 7.32, 5.40, and 23.92 cm^3/mole , calculate the pressure, at 950 K, at which Fe_3C is in equilibrium with carbon-saturated α -Fe and graphite. At what temperature, at 1 atm pressure, is carbon-saturated γ -Fe and graphite in equilibrium with cementite?

Chapter 13

Solution to 13.20



$$V_{\text{Fe}} = 7.32\text{ cm}^3/\text{mole}$$

$$V_{\text{gr}} = 5.40\text{ cm}^3/\text{mole}$$

$$V_{\text{Fe}_3\text{C}} = 23.92\text{ cm}^3/\text{mole}$$

So
$$\Delta V = 23.92 - 5.40 - 3 \times 7.32 = -3.44\text{ cm}^3/\text{mole}$$

Thus equilibrium occurs at the pressure P at which

$$2411 = 3.44(P - 1) \times \frac{8.3144}{82.057}$$

i.e. $P = 6917\text{ atm.}$

Chapter 13

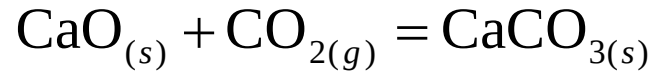
13.21 An experiment is being conducted on an equilibrated mixture of CaO and CaCO_3 contained in a closed vessel at 1200 K. The mixture is contaminated by iron in the form of hematite (Fe_2O_3). The contaminant would not be harmful to the experiment if it occurred as either wustite (FeO) or as cementite (Fe_3C). The necessary changes in the chemical form of the contaminant can be effected by admitting CO gas to the vessel. Calculate the allowable limits of p_{CO} in the vessel for the occurrence of the contaminant as

- (a) wustite and
- (b) cementite.

Chapter 13

Solution to 13.21

For

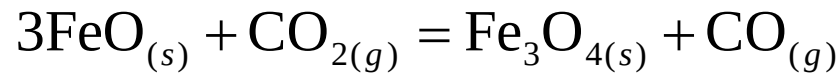


$$\Delta G_{1200 \text{ K}}^{\circ} = 3340 \text{ J} = RT \ln p_{\text{CO}_2}$$

So p_{CO_2} inside the vessel is 1.398 atm.

(a)

For



$$\Delta G_{1200 \text{ K}}^{\circ} = 4348 \text{ J} = -RT \ln \frac{p_{\text{CO}}}{p_{\text{CO}_2}}$$

So for equilibrium between FeO and Fe_3O_4 ,

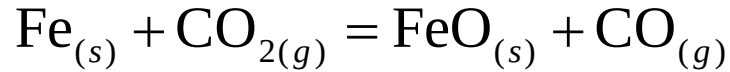
$$\frac{p_{\text{CO}}}{p_{\text{CO}_2}} = 0.647 \quad \text{or} \quad p_{\text{CO}} = 0.904 \text{ atm}$$

Thus for FeO to be stable, $p_{\text{CO}} > 0.904 \text{ atm}$

Chapter 13

Solution to 13.21

For



$$\Delta G_{1200\text{ K}}^{\circ} = -8252\text{ J} = -RT \ln \frac{p_{\text{CO}}}{p_{\text{CO}_2}}$$

So

$$\frac{p_{\text{CO}}}{p_{\text{CO}_2}} = 2.287 \quad \text{and} \quad p_{\text{CO}} = 3.196\text{ atm}$$

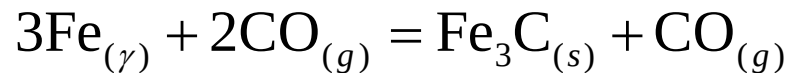
Thus for FeO to be stable, $3.196\text{ atm} > p_{\text{CO}} > 0.904\text{ atm}$

Chapter 13

Solution to 13.21

(b)

For

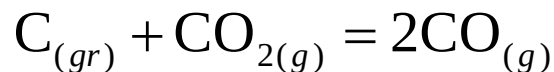


$$\Delta G_{1200\text{ K}}^\circ = 36,686\text{ J} = -RT \ln \frac{p_{\text{CO}_2}}{p_{\text{CO}}^2}$$

So

$$\frac{p_{\text{CO}_2}}{p_{\text{CO}}^2} = 2.53 \times 10^{-2} \quad \text{and} \quad p_{\text{CO}} = 7.43\text{ atm}$$

For



$$\Delta G_{1200\text{ K}}^\circ = -38,492\text{ J} = -RT \ln \frac{p_{\text{CO}}^2}{p_{\text{CO}_2}} \quad (a_{\text{C}} = 1)$$

So

$$\frac{p_{\text{CO}}^2}{p_{\text{CO}_2}} = 47.4 \quad \text{and} \quad p_{\text{CO}} = 8.14\text{ atm}$$

Thus for Fe_3C to be stable, $8.14\text{ atm} > p_{\text{CO}} > 7.43\text{ atm}$

Chapter 13

13.22 A Cu–Au alloy of $X_{\text{Cu}} = 0.5$ is being annealed at 600 °C in deoxidized argon. The argon is deoxidized by being passed over heated pure copper turnings prior to its admission to the annealing furnace. The solid Cu–Au system is virtually regular in its solution behavior, with a molar Gibbs excess free energy of mixing given by

$$G^{\text{XS}} = -28,280 X_{\text{Cu}} X_{\text{Au}} \text{ J}$$

Assuming that equilibrium is attained in the deoxidizing furnace, calculate the maximum temperature at which the deoxidizing furnace can be operated without causing oxidation of the copper in the Cu–Au alloy being annealed.

Chapter 13

Solution to 13.22

For $2\text{Cu}_{(s)} + \frac{1}{2}\text{O}_{2(g)} = \text{Cu}_2\text{O}_{(s)} \quad \Delta G_{873\text{ K}}^\circ = -101,753 \text{ J} = RT \ln p_{\text{O}_2}^{1/2} a_{\text{Cu}}^2$

So $a_{\text{Cu}}^2 \cdot p_{\text{O}_2}^{1/2} = 8.162 \times 10^{-7}$

$$\Omega = -28,280 \quad RT \ln \gamma_{\text{Cu}} = \Omega X_{\text{Cu}}^2$$

Thus
$$\ln \gamma_{\text{Cu}} = \frac{-28,280 \times 0.5^2}{8.3144 \times 873} = -0.974$$

which gives $\gamma_{\text{Cu}} = 0.378$, and $a_{\text{Cu}} = 0.378 \times 0.5 = 0.1888$

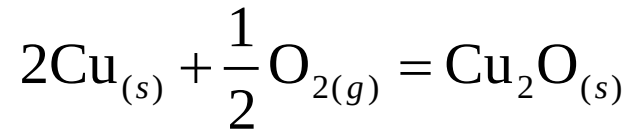
and the oxygen pressure for equilibrium between the Cu-Au alloy and Cu_2O is

$$p_{\text{O}_2} = \left[\frac{8.162 \times 10^{-7}}{0.1888^2} \right]^2 = 5.24 \times 10^{-10} \text{ atm}$$

Chapter 13

Solution to 13.22

For



$$\Delta G^\circ = -162,200 + 69.24T \text{ J} = 8.3144T \ln \left[5.24 \times 10^{-10} \right]^{1/2}$$

which gives 1026 K is the temperature at which 5.24×10^{-10} atm is the oxygen pressure required for equilibrium between Cu and Cu_2O .

The maximum temperature at which the deoxidizing furnace can be operated is thus 1026 K.

Chapter 13

13.23 In a dew-point experiment, a Cu–Zn alloy is placed in one end of an evacuated and closed tube and is heated to 900 °C. When the other end of the tube is cooled to 740 °C, Zn vapor begins to condense. Calculate the activity of Zn in the alloy relative to pure zinc.

Chapter 13

Solution to 13.23

At 740 °C

$$\ln p_{\text{Zn}}^{\circ} = \frac{-15,250}{1013} = -1.255 \ln 1013 + 21.79$$

So

$$p_{\text{Zn}, 1013 \text{ K}}^{\circ} = 0.142 \text{ atm}$$

At 900 °C

$$\ln p_{\text{Zn}}^{\circ} = \frac{-15,250}{1173} = -1.255 \ln 1173 + 21.79$$

So

$$p_{\text{Zn}, 1173 \text{ K}}^{\circ} = 0.923 \text{ atm}$$

Thus

$$a_{\text{Zn}} (\text{in the alloy}) = \frac{0.142}{0.923} = 0.154$$

Chapter 13

13.24 A crucible containing 100.0 g of silver at 1000 °C is placed in the reaction chamber of a Sieverts' apparatus. The chamber is evacuated and filled with 50 cm³ (STP) of argon, which measures the dead volume of the chamber. An external manometer reads the pressure of argon as 0.9 atm. The chamber is reevacuated and filled with oxygen, and it is found that 251.5 cm³ (STP) are required to produce a gas pressure of 0.9 atm in the chamber. Calculate the solubility of O in the Ag (as atom percent) and calculate the value of the Sieverts' law constant at 1000 °C.

Chapter 13

Solution to 13.24

Number of moles of Ar required to fill the chamber:

$$n_{\text{Ar}} = \frac{50}{82.057 \times 273} = 2.232 \times 10^{-3}$$

Number of moles of O₂ required to fill the chamber:

$$n_{\text{O}_2} = \frac{251.5}{82.057 \times 273} = 1.123 \times 10^{-2}$$

So number of moles of O₂ dissolved in liquid Ag is:

$$n_{\text{O}_2} = 1.123 \times 10^{-2} - 2.232 \times 10^{-3} = 8.995 \times 10^{-3}$$

and the weight is $8.995 \times 10^{-3} \times 32 = 0.288$ grams of O₂

Chapter 13

Solution to 13.24

Thus

$$\text{wt\% oxygen} = \frac{0.288}{0.288 + 100} \times 100 = 0.287$$

$$\text{mole fraction O, } X_{\text{O}} = \frac{2 \times 8.995 \times 10^{-3}}{2 \times 8.995 \times 10^{-3} + \frac{100}{107.87}} = 0.0190$$

So

$$\text{atom \%O} = 1.9$$

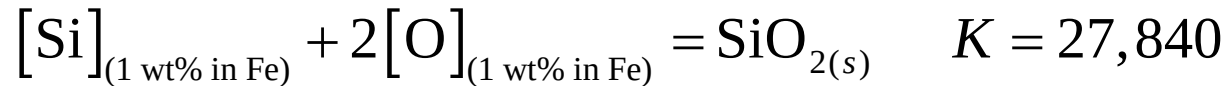
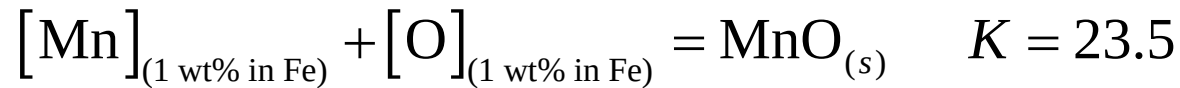
For

$$\frac{1}{2} \text{O}_{2(g)} = [\text{O}]$$

$$k = \frac{[\text{O}]}{p_{\text{O}_2}^{1/2}} = \frac{1.9}{0.9^{1/2}} = 2.0$$

Chapter 13

13.25 Silicon and manganese are commonly used together as deoxidizers for liquid steel. At 1600 °C,



The values of the equilibrium constants show that SiO_2 is considerably more stable than MnO . Why, then, is a mixture of Mn and Si more effective as a deoxidizing agent than Si alone? The activities of MnO and SiO_2 , with respect to solids as the standard states in MnO-SiO_2 melts at 1600 °C, are shown in Figure 13.37.

Chapter 13

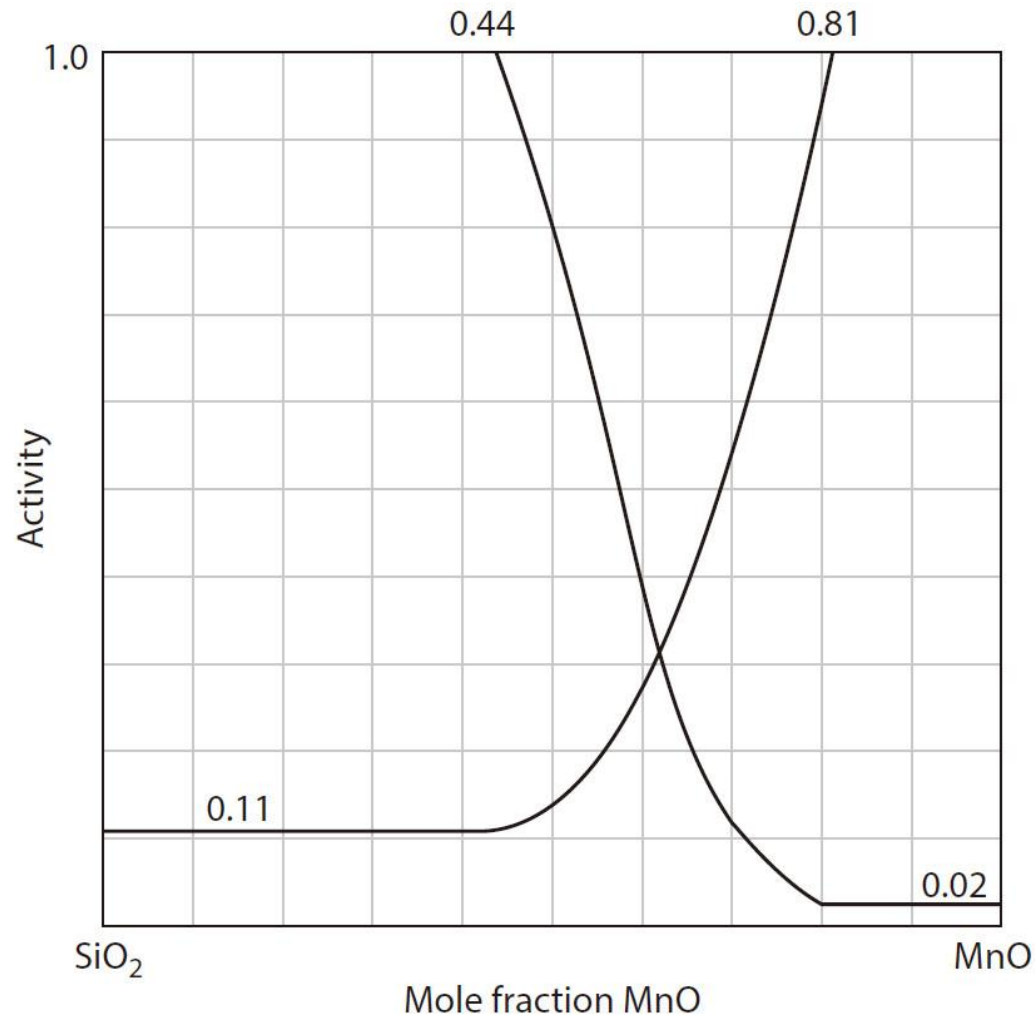


Figure 13.37 Activities in the system MnO–SiO₂ at 1600 °C.

Chapter 13

Solution to 13.25

Using Si alone produces solid SiO_2 $a_{\text{SiO}_2} = 1$ as the deoxidation product. Thus with 0.01 wt% Si, the oxygen content is

$$[\%O] = \left[\frac{a_{\text{SiO}_2}}{k_{\text{Si}} \cdot [\%Si]} \right]^{1/2} = \left[\frac{1}{27,840 \times 0.01} \right]^{1/2} = 0.06$$

Using Si + Mn produces a liquid manganese silicate as the deoxidation product, which, if saturated with MnO gives $a_{\text{SiO}_2} = 0.02$

Thus with 0.01 wt% Si, the oxygen content is

$$[\%O] = \left[\frac{0.02}{27,840 \times 0.01} \right]^{1/2} = 0.0085$$

Chapter 13

Solution to 13.25

The Mn content required to saturate with MnO is calculated as

$$[\% \text{Mn}] = \frac{a_{\text{MnO}}}{k_{\text{Mn}} \cdot [\% \text{O}]} = \frac{1}{23.5 \times 0.0085} = 5.0$$

Thermodynamics of Materials & Phase transformations

The End

